Pre−Pandemic Data Only

antiGen
EVEscape
ESM 2 (35M)
ESM 2 (650M)
BLOSUM62
Codon table

antiGen Finetunes

antiGen
antiGen as ESM 2 (35M) LM head FT
antiGen as ESM 2 (35M) full FT
antiGen as ESM 2 (35M) partial FT
antiGen as ESM 2 (650M) LM head FT
antiGen as ESM 2 (650M) full FT
antiGen as ESM 2 (650M) partial FT
antiGen as EVE full FT
antiGen as Tranception full FT

Pandemic Data in Training Window

antiGen
PSSM
PSSM Conservation
MSA Transfomer
MSA Trans. (BA.2 eval)
EVE
Tranception
Substitution matrix
Where × which

Deep Mutational Scans

antiGen
BA.1 DMS (bind)
BA.1 DMS (expression)
BA.2 DMS (bind)
BA.2 DMS (expression)
BA.2 DMS (LY−CoV1404 escape)

Ensembles

antiGen
Mean of ranks
Mean of log−probs
Mean of z−scores

Val−selected, averaged over 10 holdouts

antiGen
antiGen, val−selected (10x)
Where x which + PSSM, val−selected (10x)

all mutations, 1+ cases

de novo, 1+ cases

all mutations, 100+ cases

de novo, 100+ cases

A B C D E F G H I J K L M N O P Q R S T U V W X

P(mut. in top decile obs. by t)

Date (t, years into pandemic)